

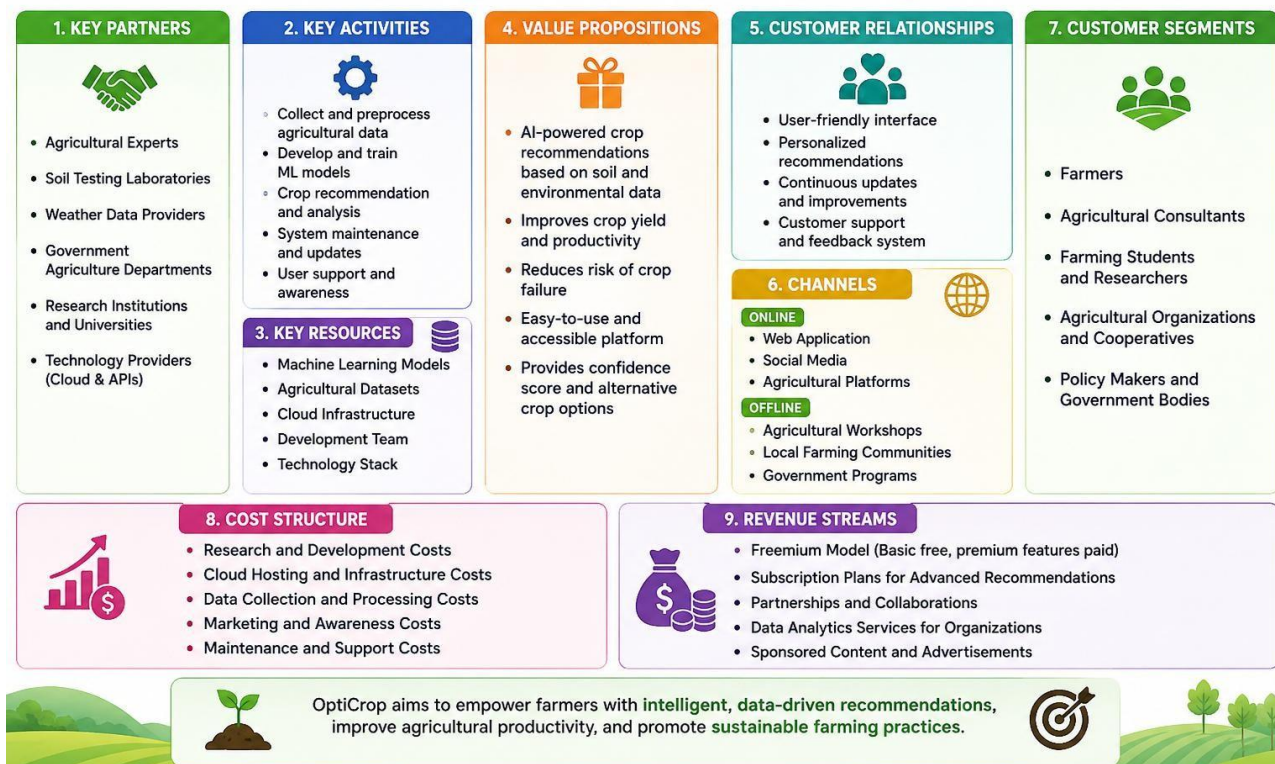
Problem-Solution Fit

Date	25 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

BUSINESS MODEL CANVAS – OPTICROP

Smart Agricultural Production Optimization Engine



Problem-Solution Fit Canvas

The Problem-Solution Fit Canvas is a strategic planning tool used to understand customer needs, identify the root causes of their challenges, and evaluate whether the proposed solution effectively addresses those problems. It helps teams analyze user behavior, emotions, limitations, and existing alternatives to ensure that the solution creates meaningful value for the target audience.

For the OptiCrop project, this canvas focuses on farmers and agricultural stakeholders who often face difficulties in selecting suitable crops due to varying soil nutrients and environmental conditions. By understanding their problems, behaviors, and expectations, the team can develop an intelligent crop recommendation system that supports data-driven agricultural decision-making.

1. Customer Segments (CS)

Identifies the primary users and beneficiaries of the solution. In OptiCrop, the main customer segments include farmers, agricultural consultants, researchers, and agricultural organizations.

2. Problems / Pains and Frequency (PR)

Describes the key challenges faced by customers and how often they occur. Farmers frequently struggle with crop selection, low productivity, and uncertainty caused by changing environmental conditions.

3. Triggers to Act (TR)

Highlights the factors that motivate customers to seek a solution. Common triggers include poor crop yields, financial losses, changing climate conditions, and the need for better farming decisions.

4. Emotions Before / After (EM)

Represents how customers feel before and after using the solution. Farmers may feel confused and uncertain before using OptiCrop, but confident and satisfied after receiving reliable crop recommendations.

5. Available Solutions – Pros and Cons (AS)

Examines existing methods used to solve the problem, such as consulting agricultural experts, relying on traditional farming practices, or using online resources. Each option has benefits but may lack personalization or accessibility.

6. Customer Limitations (CL)

Identifies barriers that prevent customers from achieving their goals. These limitations may include lack of technical knowledge, limited access to experts, budget constraints, and insufficient awareness of modern technologies.

7. Behavior and Intensity (BE)

Analyzes how customers currently behave when addressing the problem and how frequently they engage in those activities. Farmers often consult experts, seek farming advice, and rely on previous experience.

**8. Channels of Behavior (CH)**

Describes where customers search for information or support. These channels can be online, such as websites and social media, or offline, such as agricultural officers, workshops, and local farming communities.

9. Problem Root / Cause (RC)

Identifies the underlying reasons behind the customer's challenges. In agriculture, these causes include lack of data-driven tools, complex environmental factors, and dependence on traditional decision-making methods.

10. Solution (SL)

Defines the proposed solution that addresses the identified problems. OptiCrop uses Machine Learning algorithms to analyze soil nutrients and environmental conditions and recommend the most suitable crop, helping farmers improve productivity and reduce agricultural risks.